

Chapter 79 — Neoplasms of the Testis

- **Over 95% of testicular neoplasms are germ cell tumours (GCTs)**, divided into seminoma and non-seminomatous germ cell tumour (NSGCT) because their natural history and treatment differ.
- **Rare: 1–2% of cancers in adult males** in the United States; **95% arise in the testis, 5% are extragonadal**.
- **The model of a curable neoplasm:** cisplatin-based chemotherapy integrated with surgery has made GCT the paradigm of multidisciplinary cancer treatment.
- **Cure of advanced disease rose from 5–10% (pre-cisplatin) to 80–90% long-term survival** for metastatic GCT today.
- Because most patients are cured, **the central objective is minimising treatment toxicity without compromising curability**.
- **Deaths arise from inherent platinum resistance and failure to eradicate residual disease** early in the treatment course.
- **Non-GCT tumours are rare:** sex cord–stromal tumours, lymphoid and haematopoietic tumours, tumours of the collecting duct and rete testis, and adnexal tumours.

GERM CELL TUMOURS

Epidemiology

- **Burden:** in 2023 an estimated 9190 men were diagnosed with testis cancer in the United States and 470 died of it.
- **Commonest malignancy in males aged 20–40;** second commonest (after leukaemia) at 15–19.
- **Age curve:** most cases at 15–55; incidence rises steeply after puberty, **peaks at 25–35**, then declines so that men of 50–54 have the same incidence as boys of 15–19.
- **Bilateral GCT ≈2%;** most are **metachronous** (mean interval 5 years), and **histology is discordant between the two tumours in 30–50%.**
- **Geography:** highest in Scandinavia, Western Europe and Australia/New Zealand; intermediate in the United States and United Kingdom; lowest in Africa and Asia.
- **Ethnicity (United States):** incidence in non-Hispanic White men is 3.6 times that in Black men, 2.5 times that in Asian men and slightly above that in Hispanic men.
- **Incidence is rising worldwide** for both seminoma and NSGCT (United States age-adjusted rate about 3.7 per 100,000 in 1975 rising to about 6.4 in 2014); the rise among US Hispanic men has been marked, with no significant trend in other groups.
- **Stage migration** through awareness and earlier diagnosis: localised disease at diagnosis rose from 55% to 68% (1973–2014); **only about 13% present with distant metastases.**

Risk Factors

- **Five established risk factors:** White race, cryptorchidism, family history, personal history of testis cancer, and germ cell neoplasia in situ (GCNIS, also termed intratubular germ cell neoplasia, ITGCN).
- **Infertile and subfertile men** also have a higher incidence.
- **Birth-cohort effects explain the recent rise**, implying a major role for diet and environment; **early exposure to endocrine-disrupting chemicals** is associated, although no specific agent is proven.

- **Cryptorchidism**, as the examinable pair:
 - **Undescended testis: relative risk (RR) 4–6** in the affected gonad, **falling to 2–3 if orchidopexy is performed before puberty**.
 - **Contralateral descended testis: slightly increased risk** (RR about 1.7, meta-analysis).
- **Family history**: a first-degree relative raises risk substantially and lowers median age at diagnosis by 2–3 years; **RR 8–12 with an affected brother versus 2–4 with an affected father**.
- **Personal history**: **twelfold risk of contralateral GCT**, yet the **15-year cumulative incidence is only 2%**.
- **GCNIS is the precursor of most GCTs**. Exceptions: prepubertal GCTs (rarely after puberty), ovarian cystic teratomas, dermoid cysts and spermatocytic tumours.
- **GCNIS lies adjacent to invasive GCT in 80–90%** and carries a **50% risk of invasive GCT within 5 years and 70% within 7 years**.
- **Contralateral GCNIS in 5–9%** of men with GCT, **rising to about 36% with testicular atrophy or cryptorchidism**.
- **Prenatal origin**: gene expression profiling shows GCNIS develops before birth from an arrested gonocyte.
- **Microlithiasis**: in men with previous GCT, contralateral microlithiasis signals higher GCNIS risk; **in the general population its significance is unclear** — it is common, and few affected men develop GCT.

KEY POINTS — EPIDEMIOLOGY AND RISK

- GCT is the commonest solid malignancy in males aged 20–40.
- Bilateral GCT in 2%; metachronous presentation is the rule.
- Incidence highest in White and lowest in Black individuals.
- Cryptorchidism, personal or family history of GCT and GCNIS are risk factors; prepubertal orchidopexy lowers risk.
- **All postpubertal GCTs except spermatocytic tumour carry extra copies of 12p [isochromosome 12p, i(12p)]**: diagnostic for GCT in carcinoma of unknown primary and for somatic malignancy arising from teratoma.

Concept — GCT is a disease of the arrested fetal gonocyte: GCNIS forms before birth, so the risk factors (cryptorchidism, family history, previous contralateral tumour, atrophic testis) mark an abnormal developmental field, and early orchidopexy reduces the risk without abolishing it.

Histologic Classification

- **Postpubertal GCTs are GCNIS-derived or non-GCNIS-derived**; the vast majority are GCNIS-derived.
- Clinically, GCNIS-derived tumours divide into **seminoma (52–56%) and NSGCT (44–48%)**.
- NSGCT subtypes: embryonal carcinoma (EC), yolk sac tumour, teratoma and choriocarcinoma, pure or mixed; **most NSGCTs are mixed**.
- **Any NSGCT element makes the tumour an NSGCT**, however small its proportion alongside seminoma.

Germ Cell Neoplasia in Situ

- **Undifferentiated, seminoma-like germ cells lying basally within seminiferous tubules**; spermatogenesis reduced or absent.
- **Precursor of every postpubertal invasive GCT except spermatocytic tumour**.
- **No prognostic significance for relapse** when found in the orchidectomy specimen.
- **Much less frequent in paediatric GCTs**.

Seminoma

- **Commonest GCT; older patients** than NSGCT (fourth–fifth decade).
- Gross: soft, tan–white, diffuse or multinodular; necrosis focal and less prominent than in other GCTs.
- Microscopy: sheets of polygonal cells with clear cytoplasm, **nests separated by fibrovascular septa containing lymphocytes**; lymphocytic and granulomatous reactions common, with an **association with sarcoidosis**.
- **Syncytiotrophoblasts (human chorionic gonadotrophin [hCG]-positive) in about 15%**, of no clear prognostic significance.
- **Mimics**: solid-pattern EC, yolk sac tumour, Sertoli cell tumour.
- Immunohistochemistry: **CD30-negative, CD117-positive, strongly placental alkaline phosphatase (PLAP)-positive**.
- **"Anaplastic seminoma" is no longer recognised** — no biological or clinical significance.
- **Seminoma is the common precursor of NSGCT subtypes and can transform into NSGCT elements**, which shapes management of non-responders.

Spermatocytic Tumour

- Formerly "spermatocytic seminoma"; **now a distinct entity, less than 1% of GCTs**.
- **Not GCNIS-derived, not linked to cryptorchidism or bilaterality, no i(12p), never part of a mixed GCT** — sharing GCNIS independence with juvenile yolk sac tumour and teratoma.
- **Negative for OCT3/4, PLAP and glycogen** (periodic acid–Schiff, PAS); round nuclei; minimal lymphocytes; **three cell types** (small lymphocyte-like, medium eosinophilic, large mono- or multinucleated).
- **Peak in the sixth decade**.
- **Benign** — orchidectomy almost always curative; metastasis exceptionally rare.
- **Exceptions: spermatocytic tumour with sarcoma, and the anaplastic variant** — widely metastatic, chemoresistant, poor prognosis.

Embryonal Carcinoma

- Undifferentiated primitive epithelial cells with crowded pleomorphic nuclei; tan–yellow with haemorrhage and necrosis; solid, papillary, glandular-alveolar or tubular growth; syncytiotrophoblasts sometimes present.
- **Aggressive, with a high metastatic rate** — often with normal tumour markers.
- **Most undifferentiated, totipotent NSGCT**: can differentiate into any other NSGCT type, including teratoma, in the primary or at metastatic sites.
- **Presence and proportion of EC predict occult metastasis in clinical stage (CS) I NSGCT**.
- Immunohistochemistry: **AE1/AE3, PLAP and OCT3/4 positive; c-KIT negative**.

Choriocarcinoma

- **Rare, aggressive; presents with extremely high hCG and disseminated disease** — usually poor risk (stage IIIC) through hCG level and/or non-pulmonary visceral metastases.
- **Haematogenous spread** to lungs, liver and brain.
- Syncytiotrophoblasts (hCG-positive) and cytotrophoblasts; prominent haemorrhage and necrosis.
- **Haemorrhage** — spontaneous or immediately after chemotherapy starts — can be catastrophic in lung or brain.

- **Endocrine effects:** hCG shares an identical α -subunit with thyroid-stimulating hormone (TSH) and luteinising hormone (LH) → **hyperthyroidism and raised androgen production**; hyperprolactinaemia reported.

Yolk Sac Tumour

- **Pure form rare in adult testis and retroperitoneum, more common in mediastinal and paediatric GCTs;** frequent within mixed GCTs.
- Reticular network of cuboidal cells; glandular, papillary or microcystic patterns.
- **Schiller–Duval body** (resembling an endodermal sinus) in about half; **eosinophilic hyaline globules** in most.
- **Produces alpha-fetoprotein (AFP), not hCG.**

Teratoma

- Elements of at least two of the three germ layers, intermixed. **Mature versus immature has no clinical significance in postpubertal males** and is not reported.
- May contain bone, cartilage, teeth, hair and squamous epithelium ("monster tumour"); solid and cystic.
- **Markers normal or AFP mildly raised.**
- **Present in about half of adult mixed GCTs; pure teratoma uncommon.**
- **Benign histology, malignant biology:** aneuploidy, i(12p), variable proliferation; cyst fluid often contains hCG and AFP.
- **Chemoresistant** → residual masses after chemotherapy require resection; limits every chemotherapy-alone strategy.
- Unresected teratoma may **grow uncontrollably and become unresectable, undergo malignant transformation, or cause late relapse** — each potentially lethal.
- **Teratoma with somatic-type malignancy** (malignant transformation): rhabdomyosarcoma, adenocarcinoma, primitive neuroectodermal tumour; chromosome 12 abnormalities or i(12p) prove GCT origin; aggressive and chemoresistant. **Only 4% arise in the testis — most arise at metastatic sites 3–4 years after chemotherapy from unresected teratoma.**
- **Expert pathology review is worthwhile:** second-opinion review changes histologic subtype in about one-third and lymphovascular invasion status in about one-fifth, altering treatment.

GERM CELL TUMOUR SUBTYPES COMPARED

Subtype	GCNIS-derived	Markers	Immunohistochemistry	Behaviour / hallmark
Seminoma	Yes	hCG in $\approx 15\%$; never AFP	CD30–, CD117+, PLAP strongly +	Older; lymphocytic septa; can transform to NSGCT
Spermatocytic tumour	No	—	OCT3/4–, PLAP–, PAS–	Sixth decade; benign ; no i(12p); sarcomatous/anaplastic variants lethal
Embryonal carcinoma	Yes	AFP and/or hCG, or normal	AE1/AE3+, PLAP+, OCT3/4+, c-KIT–	Totipotent; metastasises with normal markers; predicts occult spread
Choriocarcinoma	Yes	Very high hCG ; no AFP	Syncytiotrophoblasts hCG+	Haematogenous; haemorrhage; thyrotoxicosis
Yolk sac tumour	Yes	AFP , not hCG	—	Schiller–Duval bodies; commoner in mediastinum and children

Subtype	GCNIS-derived	Markers	Immunohistochemistry	Behaviour / hallmark
Teratoma	Yes (postpubertal)	Normal or mild AFP	—	Chemoresistant ; growing teratoma, transformation, late relapse

- **What the table tests:** marker source by subtype. **AFP excludes pure seminoma and pure choriocarcinoma**; very high hCG points to choriocarcinoma; normal markers do not exclude EC or teratoma.
- **Trap:** spermatocytic tumour is not a seminoma variant — it is the one GCT without GCNIS or i(12p), and it is benign.

Concept — Management follows the least differentiated and most chemoresistant element: any NSGCT component makes the tumour an NSGCT, EC drives occult metastasis, choriocarcinoma drives haematogenous spread and haemorrhage, and teratoma — histologically benign but genetically malignant and chemoresistant — is the reason surgery stays integral to cure.

Initial Presentation

Signs and Symptoms

- **Painless testicular mass is the commonest presentation; any solid testicular mass on examination or imaging is malignant until proven otherwise.**
- **Acute pain** (less common) comes from rapid expansion through intratumoural haemorrhage or infarction — **commoner in NSGCT**, which is more vascular and faster-growing.
- **A history of trauma is common but incidental** — it merely draws attention to the mass. Vague scrotal discomfort or heaviness may be reported.
- **Metastases at diagnosis: about two-thirds of NSGCT versus 15% of pure seminoma**; metastatic symptoms are the presenting complaint in 10–20%.
- Metastatic features: retroperitoneal bulk (palpable mass, abdominal pain, flank pain from ureteric obstruction, back pain from psoas or nerve-root involvement, leg swelling from inferior vena cava compression, gastrointestinal symptoms); lung (dyspnoea, chest pain, cough, haemoptysis); supraclavicular node (neck mass).
- **Gynaecomastia in about 2%:** raised hCG, reduced androgen or raised oestrogen (**most often Leydig cell tumours**).
- **About two-thirds have reduced fertility**, but it is rarely the presenting complaint.

Physical Examination

- Examine both testes — relative size, consistency, intra- and extratesticular masses; **atrophy is common, especially after cryptorchidism**.
- **Any firm area within the testis is suspicious** and must be investigated.
- **A hydrocele obscuring the testis → scrotal ultrasound.**
- Examine the abdomen, **inguinal nodes (especially after previous inguinal or scrotal surgery)**, breasts, supraclavicular fossae and chest.

Differential Diagnosis

- **Epididymo-orchitis, torsion, haematoma, paratesticular neoplasm**; hernia, varicocele and spermatocele are usually distinguishable clinically.
- **A firm intratesticular mass is cancer until proven otherwise → scrotal ultrasound.**

- **Presumed epididymo-orchitis** → re-examine 2–4 weeks after completing oral antibiotics; persistent mass or pain → ultrasound.

Diagnostic Delay

- **Common, contributed to by patients and physicians; up to one-third are first misdiagnosed** as epididymitis or hydrocele.
- Young men delay through denial, ignorance or poor access; physicians may fixate on metastatic symptoms → **unnecessary laparotomy, neck dissection or mastectomy** for unsuspected GCT.
- **Delay is associated with higher stage, poorer chemotherapy response and reduced survival.** A survival penalty for long delay was shown before the cisplatin era but not within it; **unnecessary laparotomy delaying treatment beyond 30 days → more intensive chemotherapy needed.**
- **Consider GCT in any male aged 15–50 with a firm testicular mass, a midline retroperitoneal mass or a left supraclavicular mass.**

Concept — *The testis is often examined because of the metastasis, not the reverse: GCT strikes young men who delay, and misattributed metastatic symptoms lead to the wrong operation — hence the rule that a midline or left supraclavicular mass in a young man mandates testicular examination, ultrasound and markers.*

Diagnostic Testing and Initial Management

Scrotal Ultrasound

- **An extension of the physical examination** — available, inexpensive, non-invasive; 5–10 MHz transducers detect lesions of a few millimetres and separate intra- from extratesticular pathology.
- **GCT is hypoechoic** and may be multifocal: **heterogeneous echotexture → NSGCT; homogeneous → seminoma.**
- **Colour Doppler flow suggests malignancy, but its absence does not exclude GCT.**
- **Microlithiasis alone does not warrant further evaluation.**
- **Scan both testes:** bilateral GCT 2%, though synchronous bilateral tumours are rare (0.5%).
- **Advanced GCT with a normal testicular examination → ultrasound for a "burned-out" primary** (impalpable scar or calcification). GCT is among the tumours most prone to spontaneous regression — **seminoma most often.**
- **Intratesticular lesion (discrete nodule, stellate scar, coarse calcification) → radical orchidectomy,** because GCNIS and residual teratoma are frequent; normal testes on examination and ultrasound → **primary extragonadal GCT.**
- **Small (<10 mm) impalpable lesion with normal markers and no metastases is a dilemma:** most are benign (cysts, infarcts, Leydig cell nodules, small Leydig or Sertoli cell tumours) but **20–50% are small GCTs, usually seminoma; risk rises with size — about 50% below 1 cm, 80% or more at 1–2 cm.**
- Options: inguinal orchidectomy; testis-sparing inguinal exploration and excision with frozen section; or serial ultrasound with exploration of growing lesions. **Intraoperative ultrasound** localises the lesion.

Figure 79.2 — sagittal ultrasound of a left testis: multinodular hypoechoic intratesticular lesion, pure seminoma at orchidectomy.

Magnetic Resonance Imaging

- **Contrast-enhanced magnetic resonance imaging (MRI) when ultrasound is equivocal or suboptimal** — distinguishes testicular from extratesticular and neoplastic from non-neoplastic lesions.

Serum Tumour Markers

SERUM TUMOUR MARKERS COMPARED

	AFP	hCG	Lactate dehydrogenase (LDH)
Produced by	EC, yolk sac tumour	Choriocarcinoma, EC, syncytiotrophoblasts in seminoma	Non-specific (smooth, cardiac, skeletal muscle; lymphoma)
Never produced by	Seminoma, choriocarcinoma	Yolk sac tumour	—
Raised: low-stage (CS I, IIA, IIB)	50–70% of NSGCT	20–40% of NSGCT	≈20% of GCT
Raised: advanced (CS IIC, III)	60–80% of NSGCT	40–60% of NSGCT	20–60% of GCT
Seminoma	Never — raised AFP = NSGCT	≈15%	—
Half-life	5–7 days	24–36 hours	24 hours
Other causes	Hepatocellular, gastric, pancreatic, biliary and lung cancer; benign liver disease (infective, drug, alcohol, autoimmune); ataxia telangiectasia; hereditary tyrosinaemia	Liver, biliary, pancreatic, gastric, lung, breast, renal and bladder cancer; LH cross-reactivity in primary hypogonadism ; marijuana	Muscle injury; lymphoma
Distinctive points	—	Assays target the β -subunit (α shared with pituitary hormones); >5000 IU/L usually NSGCT ; hypogonadal false positive normalises 48–72 h after testosterone	LDH-1 isoenzyme commonest, tracks 12p copy number; level reflects bulk; mainly prognostic

- **What the table tests:** half-lives (to judge post-orchidectomy decline), which subtype makes which marker, and false positives. **Trap:** an hCG rise in a hypogonadal man may be LH cross-reactivity — a testosterone challenge resolves it.
- **Draw AFP, hCG and LDH before orchidectomy**, then after orchidectomy, during chemotherapy and throughout follow-up.
- **Markers never decide whether to perform orchidectomy** — normal AFP or hCG does not exclude GCT.
- **A significantly raised AFP overrides a histologic diagnosis of pure seminoma** → treat as NSGCT; borderline values need caution.
- **Medically unstable patient with markedly raised AFP and/or hCG: start chemotherapy without tissue diagnosis**, then **orchidectomy after chemotherapy** — the testis is a **sanctuary site behind the blood-testis barrier** and often retains GCT, teratoma or GCNIS.
- **Circulating microRNA (miRNA):** non-coding RNAs regulating gene expression post-transcriptionally. The **miR-371–373 (19q13) and miR-302–367 (4q25) clusters are GCT-specific**, raised in seminoma and NSGCT.
- **miR-371a-3p performs best** (sensitivity, specificity, accuracy): tracks extent, response, relapse and residual viable GCT, but **is not raised by teratoma**; intergroup trials are defining its role in low-stage and advanced disease.

Radical Inguinal Orchidectomy

- **Remove the testis and spermatic cord to the internal inguinal ring** through an inguinal incision.

- **Trans-scrotal orchidectomy or biopsy is contraindicated:** it leaves the inguinal cord and alters lymphatic drainage → local recurrence and pelvic or inguinal nodal metastasis.
- **Avoid delays beyond 1–2 weeks** — GCT grows rapidly.
- Establishes histology and pT stage, gives prognostic information, and is **curative alone in 80–85% of CS I seminoma and 70–80% of CS I NSGCT**.
- **Pathology report:** histologic type (each component and its proportion in mixed GCT), size, multifocality, local invasion (rete testis, tunica albuginea, tunica vaginalis, epididymis, cord, scrotum), primary tumour (pT) stage in the tumour, node, metastasis (TNM) system, GCNIS, **lymphovascular invasion (LVI)** and margin status.
- **Expert review recommended:** second-opinion review at a high-volume centre changed pathological stage in about a quarter (mostly LVI or cord invasion), downstaging many pT2 and pT3 tumours.

TABLE 79.2 TNM STAGING OF TESTIS TUMOR FROM THE AMERICAN JOINT COMMITTEE ON CANCER (AJCC) AND UNION INTERNATIONALE CONTRE LE CANCER (UICC) — (A) PRIMARY TUMOUR AND DISTANT METASTASIS

Category	Definition
pTx / pT0	Primary cannot be assessed / no evidence of primary (e.g. scar)
pTis	Intratubular germ cell neoplasia (carcinoma in situ)
pT1	Testis and epididymis, no LVI ; may invade tunica albuginea, not tunica vaginalis
pT2	Testis and epididymis with LVI , or through tunica albuginea into tunica vaginalis
pT3 / pT4	Invades spermatic cord / invades scrotum (± LVI)
M1a / M1b	Non-regional nodal or pulmonary metastasis / any other distant site

TABLE 79.2 (B) REGIONAL NODES: CLINICAL VERSUS PATHOLOGICAL

Clinical N (imaging)	Pathological pN (retroperitoneal lymph node dissection [RPLND] without prior chemotherapy or radiotherapy)
N1 Mass ≤2 cm, or multiple nodes none >2 cm	Mass ≤2 cm and ≤5 nodes positive, none >2 cm
N2 Mass >2 cm but ≤5 cm (single or multiple)	Mass >2–5 cm, or >5 nodes positive none >5 cm, or extranodal extension
N3 Mass >5 cm	Mass >5 cm

TABLE 79.2 (C) SERUM TUMOUR MARKERS (S) — POST-ORCHIDECTOMY

S	LDH	hCG (mIU/mL)	AFP (ng/mL)
S1 (all of)	<1.5 × upper limit of normal (N)	<5000	<1000
S2 (any of)	1.5–10 × N	5000–50,000	1000–10,000
S3 (any of)	>10 × N	>50,000	>10,000

TABLE 79.2 (D) STAGE GROUPING (COLLAPSED)

Stage	Criteria
0	pTis N0 M0 S0
IA / IB	pT1 N0 M0 S0 / pT2–4 N0 M0 S0
IS	Any pT, N0 M0, S1–3
IIA / IIB / IIC	N1 / N2 / N3, M0, S0–1
IIIA	Any N, M1a, S0–1
IIIB	N1–3 M0 S2, or M1a S2
IIC	N1–3 M0 S3, or M1a S3, or M1b any S

Data from AJCC Cancer Staging Manual, 7th ed (2010); Greene et al., AJCC Cancer Staging Manual, 6th ed (2002); Sobin and Wittekind, UICC TNM Classification, 6th ed (2002). Except pTis and pT4, pT is assigned after radical orchiectomy; Tx is used without orchiectomy.

- **What the table is for:** the **only solid-tumour TNM with a serum marker (S) category**, based on post-orchidectomy AFP, hCG and LDH.
- **Pattern:** S1 does not change stage II; **S2 or S3 with nodal disease jumps to stage IIIB/IIIC without distant metastasis**; any M1b is IIIC.
- **LVI converts an organ-confined tumour to pT2.**
- **Traps:** stage IS means raised markers with no radiological disease; clinical N is size only, whereas pN adds node count and extranodal extension — and pN cannot be applied after chemotherapy.

Testis-Sparing Surgery

- **No role for a suspected neoplasm when the contralateral testis is normal.**
- **Consider for organ-confined tumours <2–3 cm (up to 30% of testicular volume) with synchronous bilateral tumours or a solitary testis** with adequate androgen production; also for suspected benign or indeterminate lesions — **up to 80% of lesions <3 cm with symptoms for over 6 months are benign.**
- **Seldom feasible above 3 cm** — complete excision leaves too little parenchyma.
- Technique: inguinal exploration → excision → **frozen section** (reliably separates benign from malignant) → **biopsies of adjacent parenchyma for GCNIS.**
- **GCNIS → adjuvant radiotherapy ≥20 Gy** to the residual testis: prevents GCT while preserving Leydig function, but **causes permanent sterility**, and **up to 40% eventually need testosterone.**
- German Testicular Cancer Study Group: **no local recurrences with adjuvant radiotherapy, versus recurrence in most men who omitted it.**
- **Radiotherapy may be deferred to allow paternity**, with close follow-up.

Contralateral Testis Biopsy

- **Contralateral GCNIS in 5–9%; up to 36% with an atrophic testis, cryptorchidism or age under 40.**
- **Open inguinal biopsy may be considered** with these risk factors or a suspicious contralateral ultrasound.

Suspected Extragenadal Germ Cell Tumour

- **About 5% of GCTs are extragonadal:** consider GCT in any young male with a midline mass and normal testes.
- **Only one-third of metastatic GCT without a testicular mass is truly extragonadal;** most are **burned-out testicular primaries** (GCNIS in 30–50%; scar or coarse calcification on ultrasound in one-third).
- **Raised AFP and/or hCG with a normal testicular evaluation is diagnostic — biopsy is not needed** before treatment; normal markers → biopsy.
- **Poorly differentiated carcinoma with no identifiable primary:** consider extragonadal GCT with malignant transformation; **i(12p) in the biopsy supports it.**
- **Orchiectomy at some point during treatment** if the metastatic pattern fits a right- or left-sided testicular primary, or ultrasound shows a burned-out tumour.

KEY POINTS — PRESENTATION AND DIAGNOSIS

- A solid intratesticular mass in a postpubertal male is GCT until proven otherwise.
- With rare exceptions, inguinal orchiectomy with high cord ligation.
- Testis-sparing surgery only in highly selected patients in whom the preserved testis gives sufficient androgen.

- Diagnostic delay is common; about one-third are initially misdiagnosed.
- If raised before orchidectomy, re-measure markers afterwards (declining, stable or rising); pre-orchidectomy levels are not used for management decisions.

Concept — Inguinal orchidectomy is at once diagnosis, staging and, for most stage I patients, cure; every departure from it (testis sparing, chemotherapy before orchidectomy) must be justified by androgen preservation or survival, and the testis must still be dealt with because it is a chemotherapy sanctuary.

Clinical Staging

- **Clinical stage** rests on primary histology and pT stage, **post-orchidectomy markers**, and examination and imaging.
- **AJCC/UICC consensus classification (1997)** first added the marker (S) category to anatomic staging; the 2002 update made LVI in an organ-confined tumour pT2.
- **CS I = confined to testis; CS II = regional (retroperitoneal) nodes; CS III = non-regional nodes and/or visceral metastases.**

Staging Imaging Studies

- **Predictable spread:** lymphatic → retroperitoneal nodes → distant sites; **choriocarcinoma is the exception (haematogenous). The retroperitoneum is the first site in 70–80%.**
- **Right-to-left drainage:** crossover from the right landing zone is common; left → right is rare and usually reflects bulky disease.
- **Caudal (iliac, inguinal) deposits** reflect retrograde spread from bulk or, rarely, aberrant drainage.
- Lymph drains to the **cisterna chyli behind the right renal artery and right crus** (hence retrocrural nodes) → thoracic duct → posterior mediastinum → **left supraclavicular fossa.**

PRIMARY LANDING ZONES: RIGHT VERSUS LEFT TESTIS

	Right testis	Left testis
Primary landing zone	Interaortocaval nodes below the renal vessels	Para-aortic (PA) nodes (and left renal hilar)
Next echelon	Paracaval, then PA	Interaortocaval
Contralateral spread	Common	Rare — implies bulk

Clinical Staging of the Abdomen and Pelvis

- **All patients: computed tomography (CT) of abdomen and pelvis with oral and intravenous contrast** — also shows anomalies relevant to RPLND (circumaortic or retroaortic left renal vein, lower-pole renal artery, retrocaval ureter). MRI is an alternative (slower, costlier, less available).
- **Nodes enlarged on CT in 10–20% of seminomas and 60–70% of NSGCTs.**
- **The retroperitoneum is hardest to stage: 25–35% of CS I NSGCT with a "normal" CT have pathological nodes**, unchanged over four decades. No agreed size criterion; a 10-mm cut-off gives many false negatives. **CS IIA–B is overstaged in 12–40%** (negative nodes at RPLND).
- **Lower threshold in the landing zone:** a 4-mm cut-off there (10 mm elsewhere) raises sensitivity above 90% at the cost of specificity (about 50–60%) → **5–9-mm landing-zone nodes, especially anterior to the great vessels, are suspicious.**
- **Base decisions on CT within 4 weeks of treatment;** equivocal CT with normal hCG and AFP → **repeat imaging in 6–8 weeks.**

- **Fluorodeoxyglucose positron emission tomography (FDG-PET) has no routine role at diagnosis:** a prospective study of PET-negative CS I NSGCT on surveillance still found substantial relapse, and PET is barely better than CT.
- **CS II by node size:** IIA ≤ 2 cm; IIB >2 to ≤ 5 cm; IIC >5 cm.

Figure 79.3 — post-orchidectomy CT in right-sided NSGCT: 7-mm node in the primary landing zone, teratoma at RPLND.

Pathological Staging of the Abdomen and Pelvis

- **Staging RPLND without curative intent** is used for CS I–IIA NSGCT in select European centres and most laparoscopic series, to decide on chemotherapy.
- **Pathological N counts nodes:** pN1 ≤ 5 nodes, none >2 cm; pN2 >5 nodes and/or any node 2–5 cm; pN3 any node >5 cm.
- **Relapse after RPLND tracks nodal burden:** 10–30% for pN1 versus 50–80% for pN2–3.
- **pN cannot be applied after chemotherapy.**

Chest Imaging

- **Chest imaging for all before management decisions.** Thoracic metastasis without retroperitoneal disease or raised markers is uncommon (especially seminoma), so routine chest CT yields false positives that complicate care.
- **Chest radiograph first; CT if post-orchidectomy markers are raised, metastases are found on examination or abdominal CT, or the radiograph is abnormal or equivocal.** Consider chest CT in CS I NSGCT with LVI or EC predominance (haematogenous lung spread despite a clear abdomen).
- **Mediastinal or hilar nodes without retroperitoneal disease → suspect lymphoma or sarcoidosis;** confirm GCT by mediastinoscopy and biopsy before chemotherapy.
- **No routine bone scan or brain imaging — except brain imaging when hCG exceeds 10,000 IU/L** (metastatic choriocarcinoma, which favours the brain).

Serum Tumour Markers (Staging)

- **Post-orchidectomy AFP, hCG and LDH for everyone:** confirm decline according to half-life when levels were raised.
- **New or rising markers after orchidectomy = metastatic disease → induction chemotherapy.**
- **Slow decline (not by half-life) with negative imaging → monitor** until normal or rising.
- **Stable, slightly raised AFP or hCG → exclude other causes** before acting.
- **Use markers measured within 4 weeks of treatment.**

Prognostic Classification of Advanced Germ Cell Tumour

- **International Germ Cell Cancer Collaborative Group (IGCCCG), 1997:** pooled analysis of 5202 men with advanced NSGCT treated with platinum (1975–1990) identified **AFP, hCG and LDH at the start of chemotherapy, non-pulmonary visceral metastases (NPVM) and primary mediastinal NSGCT** as independent predictors; in 660 advanced seminomas **only NPVM** mattered.
- **Assign an IGCCCG group to every metastatic GCT at diagnosis** to choose chemotherapy. It **does not apply to relapse** and uses **post-orchidectomy markers at the start of chemotherapy**, not pre-orchidectomy values.
- **No poor-risk category exists for seminoma.**
- **Survival has improved since:** a meta-analysis of NSGCT treated after 1989 gave 5-year survival of 94%, 83% and 71% (good, intermediate, poor) — especially better for poor risk, reflecting better therapy and experience.

- **TNM versus IGCCCG:** TNM stages both seminoma and NSGCT with NPVM as IIIC, whereas IGCCCG calls the former intermediate and the latter poor risk → **IGCCCG is preferred for prognosis and choice of chemotherapy.**

TABLE 79.3 INTERNATIONAL GERM CELL CANCER COLLABORATIVE GROUP RISK CLASSIFICATION FOR ADVANCED GERM CELL TUMOR

Risk	Non-seminoma	Seminoma
Good	Testicular/retroperitoneal primary and no NPVM and all of: AFP <1000 ng/mL, hCG <5000 IU/L (1000 ng/mL), LDH <1.5 × N 56%; 5-year progression-free survival (PFS) 89%, survival 92%	Any primary site and no NPVM; normal AFP, any hCG, any LDH 90%; 5-year PFS 82%, survival 86%
Intermediate	Testicular/retroperitoneal primary and no NPVM and any of: AFP 1000–10,000, hCG 5000–50,000 IU/L, LDH 1.5–10 × N 28%; 5-year PFS 75%, survival 80%	Any primary site with NPVM ; normal AFP, any hCG, any LDH 10%; 5-year PFS 67%, survival 72%
Poor	Mediastinal primary, or NPVM, or any of: AFP >10,000, hCG >50,000 IU/L (10,000 ng/mL), LDH >10 × N 16%; 5-year PFS 41%, survival 48%	No patients

PFS, progression-free survival; N, upper limit of normal. Data from International Germ Cell Cancer Collaborative Group, J Clin Oncol 1997.

IGCCCG SURVIVAL: TABLE VALUES VERSUS OTHER FIGURES QUOTED IN THE CHAPTER

Group	Table 79.3 (IGCCCG 1997 cohort)	Figure quoted elsewhere in text	Reason for difference
NSGCT good	5-year survival 92%	91–94% (with BEP×3 [bleomycin, etoposide, cisplatin] or EP×4 [etoposide, cisplatin]); 94% (post-1989 meta-analysis)	Later cohorts and pooled data
NSGCT intermediate	5-year survival 80%	79% ("at the time categories were defined"); 83% (post-1989)	Rounding of same cohort; later cohorts
NSGCT poor	5-year survival 48%	71% (post-1989)	Improved therapy and experience
Seminoma good	5-year survival 86%	91% (advanced seminoma series)	Different series, not the IGCCCG derivation cohort
Seminoma intermediate	5-year PFS 67%, survival 72%	PFS 75%, survival 79% (with BEP×4)	Different series; the IGCCCG figures are the reference values

- **What the table is for:** selecting the regimen and number of cycles for metastatic GCT.
- **Pattern:** NSGCT risk is driven by markers, primary site and NPVM; seminoma risk by NPVM alone — markers and mediastinal primary do not count in seminoma.
- **Traps:** mediastinal primary NSGCT is automatically poor risk, whereas mediastinal seminoma can be good risk; the classification uses **post-orchidectomy** markers at the start of chemotherapy and is **not valid at relapse**.

Sperm Cryopreservation

- Up to half have oligospermia at diagnosis and 10% are azoospermic.
- **Germinal epithelium is exquisitely sensitive to platinum and radiation:** virtually all men become azoospermic after chemotherapy; of those normal at baseline, **about half recover by 2 years and 80% by 5 years.**
- **Recovery after radiotherapy for seminoma takes 2–3 years or more.**
- **Full bilateral template RPLND without nerve sparing → ejaculatory dysfunction in 80% or more.**

- **Offer sperm banking before RPLND, chemotherapy or radiotherapy** to men undecided about or planning paternity — before or after orchidectomy.

KEY POINTS — STAGING

- Spread is predictable: primary → retroperitoneal nodes → distant sites.
- Landing zone: left = PA and left renal hilar nodes; right = interaortocaval and paracaval nodes.
- CT is optimal for the retroperitoneum, but with a 1-cm cut-off false negatives occur in 25–35% of CS I NSGCT and 14–20% of CS I seminoma.
- Rising post-orchidectomy markers indicate metastatic GCT → chemotherapy.
- IGCCCG classification gives prognosis and dictates chemotherapy.
- Offer sperm cryopreservation to all before RPLND, chemotherapy or radiotherapy.

Concept — Staging is anatomy plus biology: the lymphatic map (side-specific landing zones, right-to-left crossover, cisterna chyli to left supraclavicular fossa) tells you where to look and why CT under-stages, while post-orchidectomy markers read against their half-lives are the only direct signal of occult systemic disease — which is why the marker category, not node size alone, governs stage and risk group.

Treatment

Therapeutic Principles

- **Rapid growth plus near-universal curability → rapid diagnosis, staging and treatment**, so that no patient dies unnecessarily or needs toxicity that earlier care would have avoided.
- **An aggressive approach is justified:** chemotherapy is given despite low white cell or platelet counts; **cisplatin is given even with moderate-to-severe renal impairment; all residual disease is resected after chemotherapy, even at multiple sites.** Young, fit patients tolerate this.
- **Markers steer management:** raised after orchidectomy → chemotherapy; rising during or after therapy → refractory or relapsed disease; levels at the start of chemotherapy set regimen and duration.
- **Centralise care:** GCT is rare, algorithms are nuanced, RPLND evidence comes from a few high-volume surgeons, and most US urology residents perform two or fewer RPLNDs; **survival is better at high-volume centres** → treat there, with RPLND by experienced surgeons.

Contrasting Seminoma and Nonseminoma Germ Cell Tumour

SEMINOMA VERSUS NSGCT — THE GOVERNING CONTRAST

	Seminoma	NSGCT
Natural history	Indolent; orderly lymphatic spread before haematogenous	Aggressive
CS I / II / III at diagnosis	85% / 10% / 5%	≈33% / 33% / 33%
Occult metastasis in CS I	10–15%	25–35%
Systemic relapse after retroperitoneal treatment	1–4% (after radiotherapy)	≈10% (after RPLND)
Markers	Less often raised, lower levels; not used in IGCCCG	Central to staging, risk and response
Radiosensitivity	Exquisite — low doses curative; standard option for CS I and IIA–B	No role , except brain metastases
Platinum sensitivity	Greater	High
Share of advanced GCT	Only 10% (despite 52–56% of all GCT)	Majority

	Seminoma	NSGCT
IGCCCG	No poor risk; >90% good	56% good
Metastatic teratoma	Generally not a concern	Frequent → post-chemotherapy surgery
Transformation	NSGCT elements in 10–15% of relapses; 30% at autopsy of seminoma deaths	—

- **Metastatic teratoma drives NSGCT algorithms:** frequent post-chemotherapy surgery (PCS) for advanced disease, and a preference for RPLND in low-stage disease with low risk of distant spread; **outcome of metastatic teratoma depends on completeness of resection.**
- **Seminoma's capacity to transform** matters in non-responders and in relapse after radiotherapy.
- **"Double therapy"** (surgery plus chemotherapy) is minimised where possible, but mixed NSGCT often needs chemotherapy for sensitive elements and surgery for teratoma; **integration of chemotherapy and PCS is a major reason cure rates have risen.**
- **Chemotherapy, radiotherapy and CT imaging raise lifetime risks of secondary malignant neoplasm (SMN) and cardiovascular disease,** whereas **RPLND by experienced surgeons has the more favourable long-term toxicity profile.**

KEY POINTS — SEMINOMA VERSUS NSGCT

- Seminoma is indolent, with less metastatic disease.
- No poor-risk category for metastatic seminoma.
- Seminoma is more sensitive to radiotherapy and platinum chemotherapy.
- hCG is raised in only 15% of metastatic seminoma; markers do not guide seminoma treatment.
- Metastatic teratoma is less of a concern in seminoma but must be considered in non-responders.

Concept — Two properties split every algorithm: seminoma's radiosensitivity and near-absence of metastatic teratoma permit radiotherapy and observation of residual masses, whereas NSGCT's chemoresistant teratoma makes surgery obligatory — so avoiding double therapy is always subordinate to eradicating every element.

Germ Cell Neoplasia In Situ

- Diagnosed on biopsy for infertility, contralateral biopsy in GCT, or within a testis-sparing specimen; **treated because of the high risk of invasive GCT.**
- **Options: radical orchiectomy (most definitive), low-dose radiotherapy (≥ 20 Gy), close observation** — chosen by wish for paternity, state of the contralateral testis and wish to avoid testosterone replacement.
- **≥ 20 Gy gives local control similar to orchiectomy while preserving endocrine function,** because **Leydig cells are relatively radioresistant compared with germinal epithelium;** but **up to 40% eventually need testosterone,** so monitor androgen production. **Doses < 20 Gy → recurrent GCNIS.**
- **Normal contralateral testis and wish for paternity → orchiectomy,** because radiotherapy scatter impairs contralateral spermatogenesis.
- **Subnormal semen still adequate for assisted reproduction → surveillance with serial ultrasound,** deferring treatment until pregnancy or GCT; alternatively testicular exploration, sperm harvest and cryopreservation, then orchiectomy and testosterone.
- **GCNIS with planned cisplatin chemotherapy:** chemotherapy reduces but does not eliminate risk (about one-fifth by 5 years, nearly half by 10 years) → **low-dose radiotherapy after chemotherapy, or biopsy 2 years or more after chemotherapy** and treat only if GCNIS persists.

KEY POINTS — GCNIS

- GCNIS is the precursor of GCT, with a 50% risk of invasive GCT within 5 years.
- Radical orchidectomy and low-dose (≥ 20 Gy) radiotherapy are effective.

Concept — GCNIS treatment trades three testicular functions — germ cells (fertility), Leydig cells (androgen) and cancer control — by exploiting the differential radiosensitivity of germinal epithelium and Leydig cells.

Nonseminoma Germ Cell Tumours

Clinical Stage I Nonseminoma Germ Cell Tumour

- **About one-third of NSGCT patients are CS I with normal post-orchidectomy markers;** long-term survival with surveillance, RPLND or primary chemotherapy approaches 100%, hence the controversy.
- **Occult metastases (retroperitoneal or distant) in only 20–30% →** any post-orchidectomy intervention overtreats the 70–80% with disease confined to the testis.
- **Most centres use a risk-adapted approach;** some offer surveillance to all regardless of risk.

Risk assessment

- **LVI and a predominant EC component are the main predictors of occult metastasis;** "EC predominance" is defined variously as 45–90% EC.
- In older series, occult metastasis ran at 45–90% with LVI and 30–80% with EC predominance; **with neither factor, risk is under 20%.**
- Other predictors: advanced pT stage, absence of mature teratoma, absence of yolk sac tumour, any EC, MIB-1 staining, tumour size, age. A pooled analysis ranked **LVI (odds ratio about 5), MIB-1 staining >70% (about 5) and EC predominance (about 3)** strongest.
- **Factor in the CT:** landing-zone nodes above 5–9 mm are suspicious, since a 1-cm cut-off misses many.
- **CS I low/high risk (by LVI and EC) is not the IGCCCG classification,** which applies only to metastatic disease. Only one CS I model has been prospectively validated, and none incorporates staging CT.
- **Contemporary risks are lower:** prospective series put occult metastasis with LVI or EC predominance at 35–55% (relapse on surveillance about 50% with LVI, one-third with EC predominance); in the Albers RPLND trial only 18% were node-positive although 42% had LVI — attributed to closer CT scrutiny and stage migration.

Surveillance

- **Rationale:** 70–80% are cured by orchidectomy, and relapses are salvaged by good-risk chemotherapy; survival matches RPLND and primary chemotherapy → **a standard option** that confines treatment to those proven to need it.
- **Disadvantages:** highest relapse risk; follow-up beyond 5 years; SMN risk from repeated CT; relapse needs more intensive treatment than upfront therapy would have.
- **Most-used approach in the United States** for CS IA (about three-quarters) and CS IB (about half).
- Pooled series of over 3000 men: **mean relapse about 28%, cancer-specific mortality about 1%.**
- **Over 90% of relapses occur within 2 years;** late relapse (>5 years) in up to 1% (up to 5% in some reports).
- **65–75% of relapses are retroperitoneal,** with or without raised markers. Most need induction chemotherapy (bulky >3 cm nodes, raised markers or distant disease); **normal markers with nonbulky (<3 cm) retroperitoneal relapse → RPLND is an option.**

- **Schedule:** no schedule proven superior; intensive in years 0–2, less frequent in years 3–5, and beyond 5 years for late relapse (whether with CT is controversial). CT use ranges from 2 to 13 or more scans in 5 years.
- **Medical Research Council (MRC) randomised trial: 2 versus 5 CT scans in years 1–2** → no difference in survival, IGCCCG group or stage at relapse.
- **Non-compliance with surveillance: 35–80%.**
- **Relapses are found by CT or markers; chest radiograph and examination rarely detect them.** With LVI, relapse is more often marker-detected; without LVI, more often CT-detected.
- **Late relapses on surveillance are treated as successfully as early ones.**

Retroperitoneal lymph node dissection

- **Rationale for RPLND:**
 - retroperitoneum is the commonest site of occult disease, with low associated systemic risk
 - **teratoma in 15–25% of occult retroperitoneal metastases** (chemoresistant)
 - low abdominal–pelvic recurrence after full bilateral template → **no routine surveillance CT**
 - high cure with RPLND alone for low-volume malignancy (pN1) and teratoma (pN1–3)
 - **≥75% avoid chemotherapy** if adjuvant therapy is reserved for pN2–3
 - relapses salvaged by good-risk chemotherapy
 - low short- and long-term morbidity with nerve sparing by experienced surgeons
- **In low-stage NSGCT the therapeutic focus is the retroperitoneum**, and RPLND gives the best control with the least serious long-term morbidity.
- **Disadvantages:** major abdominal surgery for all; needs experienced surgeons (not deliverable everywhere); **highest rate of double therapy**. US use is falling (about 15% for CS IA and IB).
- Pathological stage II in 19–28%; **66–81% of these are cured by RPLND alone**; cancer-specific survival (± adjuvant chemotherapy) approaches 100% with negligible late relapse; **retroperitoneal recurrence under 2%.**
- **Teratoma only in the retroperitoneum** → low relapse whatever the pN stage; **no adjuvant chemotherapy.**
- **Full bilateral template: abdominal–pelvic recurrence <2% and antegrade ejaculation >90% with nerve sparing** → standard of care for primary RPLND; **mandatory with suspicious nodes (CT or intraoperative) or somatic-type malignancy in the primary**; a modified template is acceptable for clinically negative nodes.
- **Modified templates:** right-sided may omit PA nodes below the inferior mesenteric artery (IMA) — omitting PA nodes above the IMA is controversial; left-sided may omit paracaval, precaval and retrocaval nodes — omitting interaortocaval nodes is controversial.
- **RPLND cures 60–90% of pN1 and up to 100% of teratoma-only disease; relapse with pN2–3 exceeds 50%.**
- **Two adjuvant cycles (BEP×2 or EP×2) cut relapse to 1% or less.** In the randomised adjuvant trial (Williams), relapse was 6% with adjuvant chemotherapy versus 49% with observation, **with no overall survival difference** → **adjuvant chemotherapy and observation are both acceptable** for pathological stage II; counsel on the trade-off.

Primary chemotherapy

- **Primary chemotherapy = chemotherapy after orchidectomy for CS I**, as distinct from adjuvant chemotherapy after RPLND for pathological stage II.

- **Rationale:** two cycles eradicate micrometastases in the adjuvant setting, and 20–25% need chemotherapy despite RPLND (adjuvant or at relapse). Goal: minimise relapse and avoid RPLND and later induction chemotherapy.
- **Advantages:** **lowest relapse of any single modality**; deliverable in community centres.
- **Disadvantages:** **does not treat retroperitoneal teratoma** (risk of chemoresistant or late relapse); long-term retroperitoneal CT surveillance still required; everyone is exposed to late toxicity, and **no safe lower limit** of exposure is known.
- Used far less in the United States (about 13% CS IA, 37% CS IB) than in Europe.
- **With LVI and/or EC predominance, relapse falls from 30–60% to about 2–3%.** Pooled series: relapse about 2.4%, mortality about 0.45% — but **about 18% of relapsing men died**: relapses after chemotherapy are chemoresistant, especially after non-standard regimens. Relapse after surveillance or RPLND is chemo-naïve and almost always curable. **Overall cure (>99%) is similar** across approaches.
- **Relapses after primary chemotherapy are almost all retroperitoneal → CT surveillance of abdomen and pelvis.**
- Many European centres prefer primary BEP because RPLND there is mainly a staging procedure.
- **Guidelines now recommend BEP×1:** over 1000 patients, relapse about 2.3%, mortality about 0.2% — not appreciably different from BEP×2.

CLINICAL STAGE I NSGCT — SURVEILLANCE VERSUS RPLND VERSUS PRIMARY CHEMOTHERAPY

	Surveillance	RPLND	Primary chemotherapy (BEP×1)
Men treated	Only those who relapse	All (major surgery)	All (chemotherapy)
Relapse	Highest (≈28% overall)	Intermediate; retroperitoneal <2%	Lowest (≈2–3%)
Where relapse occurs	Retroperitoneum 65–75%	Lungs / mediastinum	Retroperitoneum
Relapse salvage	Chemo-naïve; near-certain	Chemo-naïve; near-certain	Chemoresistant; poorer salvage
Abdominal CT follow-up	Intensive, >5 years	Not routinely needed	Required long term
Retroperitoneal teratoma	Left in situ	Removed	Left untreated
Double therapy	Low	Highest	—
Late toxicity	CT radiation	Most favourable (expert surgeon)	Chemotherapy late effects
Long-term survival	≈100% for all three		

Treatment selection

- **No randomised trial compares the standard approaches.** The **Albers (German Testicular Cancer Study Group) phase III trial** — BEP×1 versus unilateral modified-template RPLND (BEP×2 if pathological stage II) — favoured BEP×1 for relapse (2-year PFS 99% vs 92%; hazard ratio 0.13) with **no GCT deaths in either arm**. RPLND was spread across 61 centres, and in-field recurrence was 11% after node-negative RPLND — inexperience and unilateral templates. **Criticised as comparing two non-standard treatments.**
- → **RPLND only by an experienced surgeon with a full bilateral (or accepted modified) template; otherwise surveillance or primary chemotherapy.**
- **No single option should be universally recommended;** weigh toxicity, need for further treatment, surveillance intensity and patient preference.

- **Guidelines:** low risk → surveillance; high risk → surveillance, RPLND or primary chemotherapy.
- **Decision analysis** (cancer outcomes, toxicity, patient preferences): surveillance maximises quality-adjusted survival when relapse risk is below 33–37%; active treatment is favoured above 46–54%.

Concept — CS I NSGCT is a question of who carries the 20–30% occult disease: all three options give equal survival and differ only in how many men are treated needlessly versus how hard relapses must be treated — hence risk adaptation by LVI and EC, with a threshold of about one-third relapse risk.

Clinical Stage IS Nonseminoma Germ Cell Tumour

- **CS IS = raised post-orchidectomy markers without clinical or radiographic metastasis.**
- After primary RPLND, 37–100% still needed chemotherapy → **consensus: treat like CS IIC–III with induction chemotherapy**; cancer-specific survival >90%.
- **Slightly raised, stable markers may be false positives** — interpret cautiously.

Clinical Stage IIA and IIB Nonseminoma Germ Cell Tumour

- **RPLND (± adjuvant chemotherapy) and induction chemotherapy (± post-chemotherapy RPLND) are both accepted**, with survival above 95%; no randomised comparison exists.
- **Non-randomised comparisons:** a prospective multicentre study found no difference in recurrence or survival; a single-institution comparison found fewer recurrences with induction chemotherapy but near-100% cancer-specific survival with both, **fewer chemotherapy cycles with RPLND (mean 1.4 vs 4.2), and about half of RPLND patients avoiding chemotherapy.**

CLINICAL STAGE IIA–B NSGCT — RPLND VERSUS INDUCTION CHEMOTHERAPY

	Primary RPLND	Induction chemotherapy
For	13–35% are pN0 (CT false positive) and avoid chemotherapy; ≈30% have retroperitoneal teratoma; cancer-specific survival 98–100%; 10–52% avoid any chemotherapy; ejaculation preserved in 70–90%	60–78% achieve complete response and avoid PCS; deliverable in community centres; cancer-specific survival 96–100%
Against	≥48% need further therapy; 13–15% have persistent disease needing full induction; high-quality RPLND not universally available	All exposed to late chemotherapy toxicity; without post-chemotherapy RPLND, risk of chemorefractory relapse

- **Indeterminate nodes on CT with otherwise low risk → observe briefly** to clarify, given the 13–35% false-positive rate.
- **Considerations:** risk of occult systemic disease, risk of retroperitoneal teratoma, short- and long-term morbidity, and double therapy — **the last is least important**, though it has swayed opinion; cure often needs both chemotherapy and surgery.
- **Predictors of systemic relapse after RPLND:** raised post-orchidectomy AFP or hCG, and nodes >3 cm → **consensus: raised markers or nodes >3 cm → induction chemotherapy.**
- **Retroperitoneal teratoma in about 20% of CS IIA–B, 30–35% if teratoma is in the primary.** Microscopic teratoma may lie dormant for life or grow slowly (detectable and resectable), but **growing teratoma syndrome, malignant transformation and late relapse** are its serious sequelae → **RPLND first when teratoma is likely and systemic risk is low (normal markers, nodes <3 cm).**

Concept — Markers and bulk predict systemic disease; teratoma predicts the need for surgery — raised markers or nodes over 3 cm point to chemotherapy, whereas marker-negative nonbulky disease, especially with teratoma in the primary, points to RPLND.

Clinical Stage IIC and III Nonseminoma Germ Cell Tumour

- **Cisplatin-based induction chemotherapy** for CS IIC–III — and for CS IS and CS IIA–B with raised post-orchidectomy markers; regimen and cycles by IGCCCG group.

- **Cisplatin is the single most important advance:** before it, complete response was 10–20% and cure 5–10%; **long-term cure is now 80–90%** of metastatic GCT.
- **Indiana University cisplatin–vinblastine–bleomycin (PVB×4), 1970s:** complete response 74%, over 70% long-term survivors.
- Etoposide salvaged some PVB failures → **PVB×4 versus BEP×4:** equal survival (2-year about 80%) but **less neuromuscular toxicity with BEP** → **BEP became standard.**

Chemotherapy for good-risk NSGCT

- Aim after BEP×4 became standard: **less toxicity for good risk, better outcomes for intermediate and poor risk.**
- **BEP×3 is non-inferior to BEP×4** → standard for good risk; **carboplatin was inferior to cisplatin in every randomised comparison.**
- **Omitting bleomycin** (pulmonary fibrosis, Raynaud phenomenon) is one of the few remaining controversies: trials trend towards bleomycin without a significant survival difference → **BEP×3 or EP×4 (full-dose etoposide) accepted for good risk**, 5-year survival 91–94% (trial detail in the table below).

Chemotherapy for intermediate- and poor-risk NSGCT

- **BEP×4 standard since 1987.**
- **VIP×4 (etoposide, ifosfamide, cisplatin)** matched BEP×4 for survival but caused more toxicity → BEP×4 remains standard; because ifosfamide can replace bleomycin, **VIP×4 is substituted when pulmonary function is compromised or extensive chest surgery is expected.**
- **High-dose chemotherapy (HDCT) with autologous stem-cell rescue:** rationale — dose escalation may overcome resistance; **carboplatin replaces cisplatin because cisplatin's nephrotoxicity and neuropathy limit dose.** First-line HDCT failed in two randomised trials.
- **Slow AFP/hCG decline** (longer time to normalisation and longer half-lives) is adverse; intensifying chemotherapy for slow decliners improved PFS but not survival → **BEP×4 remains standard first line.**

CHEMOTHERAPY REGIMENS IN GCT

Regimen	Agents	Line / setting	Key point
BEP	Bleomycin, etoposide, cisplatin	First line: ×3 good risk; ×4 intermediate/poor; ×1 primary for CS I; ×2 adjuvant	Standard since superseding PVB
EP	Etoposide, cisplatin	×4 good risk; ×2 adjuvant	Bleomycin-free; EP×3 inferior
PVB	Cisplatin, vinblastine, bleomycin	Historical first line	Replaced by BEP (neuromuscular toxicity)
VIP	Etoposide, ifosfamide, cisplatin	First line ×4 if lung function poor / major chest surgery planned; second line	More haematological and urological toxicity
VeIP	Vinblastine, ifosfamide, cisplatin	Second line after prior etoposide	Standard second line
TIP	Paclitaxel, ifosfamide, cisplatin	Second line; best results in late relapse	Standard second line
HDCT	Carboplatin–etoposide ± cyclophosphamide, ifosfamide, paclitaxel or thiotepa; stem-cell rescue	Second/third line at specialised centres	Not superior in three randomised trials
Carboplatin (single agent)	Carboplatin (Calvert-dosed)	Primary chemotherapy, CS I seminoma	Inferior to cisplatin in advanced disease

Regimen	Agents	Line / setting	Key point
Dose-dense	Paclitaxel, bleomycin, etoposide, cisplatin, oxaliplatin → cisplatin, ifosfamide, bleomycin	Slow marker decline after BEP×1 (trial)	PFS gain, no survival gain
Later-line agents	Gemcitabine, paclitaxel, oxaliplatin, irinotecan	After second-line failure	Combinations

SIGNATURE TOXICITIES OF THE COMPONENT DRUGS

Drug	Signature toxicity (as given in the chapter)
Cisplatin	Nephrotoxicity, peripheral neuropathy, ototoxicity (hearing loss, tinnitus)
Bleomycin	Pulmonary toxicity including fibrosis; Raynaud phenomenon
Ifosfamide (VIP)	High-grade haematological and urological toxicity
Vinblastine (PVB)	Neuromuscular toxicity
Carboplatin	Less neuro-, oto- and nephrotoxicity than cisplatin; haematological toxicity

FIRST-LINE CHEMOTHERAPY TRIALS — ADVANCED NSGCT

Question	Comparison	Population	Result	Verdict
Etoposide for vinblastine	PVB×4 vs BEP×4	Advanced GCT	Equal survival; less neuromuscular toxicity	BEP standard
Fewer cycles	BEP×3 vs BEP×4 (US; European)	Good risk	Equal PFS and survival	BEP×3 good risk
Carboplatin for cisplatin	Carboplatin vs cisplatin regimens	Good risk	Cisplatin superior in all	Keep cisplatin
Omit bleomycin	EP×3 vs BEP×3	Good risk	EP×3 inferior	Reject EP×3
Omit bleomycin	EP×4 (reduced-dose etoposide) vs BEP×4 (European)	Good risk	More complete responses with BEP; equal survival	—
Omit bleomycin	EP×4 (full-dose etoposide) vs BEP×3 (French)	Good risk	No significant difference in relapse or survival	EP×4 accepted
Ifosfamide for bleomycin	VIP×4 vs BEP×4 (US; European closed early)	Intermediate/poor	Equal survival; more haematological and urological toxicity with VIP	BEP×4; VIP if lungs at risk
High-dose first line	BEP×2 + high-dose carboplatin–etoposide–cyclophosphamide ×2 vs BEP×4 (Motzer); smaller second trial	Intermediate/poor	No difference in durable complete response or survival; more toxicity	Reject
Marker-guided intensification	Dose-dense vs BEP×3 more (Fizazi; French)	Slow marker decline after BEP×1	3-year PFS 59% vs 48% (hazard ratio 0.66); no survival gain	BEP×4 remains

- **Pattern across the trials:** de-escalation for good risk succeeded (BEP×3, EP×4); every attempt to intensify for poor risk (ifosfamide, high dose, dose-dense) failed to improve survival. **Trap:** carboplatin is acceptable only as single-agent primary therapy in CS I seminoma, never in place of cisplatin for metastatic disease.

Management of post-chemotherapy residual masses in NSGCT

- **Restage** with markers and imaging of chest, abdomen and pelvis (and other initial sites). Response categories:
 - **complete response (CR):** markers normal and residual masses ≤1 cm
 - partial remission, marker-negative (persistent radiographic tumour)
 - partial remission, marker-positive

- progression — the last two groups (5–15%) receive second-line (salvage) chemotherapy
- **Residual mass >1 cm (38–68% of patients) → PCS: clear consensus.** After CR, practice is controversial: some guidelines observe, others resect if the pre-chemotherapy mass exceeded 3 cm.
- **Why resect: persistent GCT elements (teratoma or viable malignancy) in 50% or more** — histology by setting in the table below.
- **Completely resected viable malignancy: 5-year survival 45–77%;** left unresected it relapses, and only 25–35% achieve durable remission with second-line chemotherapy.
- **Teratoma at metastatic sites in 15% or more;** unresected it risks growing teratoma syndrome, transformation or late relapse; **long-term survival 75–90% after resection of residual teratoma.**
- **In-field retroperitoneal relapse <2% after full bilateral template** → abdominal imaging surveillance largely unnecessary.
- **Non-GCT malignancy (malignant transformation) in 6–8% of PCS specimens; rhabdomyosarcoma commonest;** i(12p) or chromosome 12 abnormalities confirm GCT origin. Resistant to GCT chemotherapy → **complete resection gives 50–66% survival**, incomplete resection is usually fatal; histology-specific chemotherapy gave partial responses in selected patients.
- **Necrosis only → relapse ≤10%.**
- **Predictors of necrosis:** no teratoma in the primary, greater percentage shrinkage, small residual size — yet **models carry a consistent ≈20% false-negative rate**, so necrosis cannot be predicted well enough to omit PCS. **No teratoma in the primary does not exclude retroperitoneal teratoma.**
- **FDG-PET has no role for NSGCT residual masses:** teratoma is not FDG-avid, and PET predicted histology no better than CT or markers.
- **After complete response (26–62%):** residual masses ≤20 mm still harbour teratoma or viable malignancy (table), so size cannot exclude disease. Observation of complete responders gave 92% survival at 5–10 years (Indiana), and recent series of residual masses <1 cm observed without PCS reported relapse 4–10% and cancer-specific survival 97–100% — **but these were mostly good risk without teratoma in the primary, i.e. highly selected;** complete responders are a small minority.
- **Multiple sites (about one-third; retroperitoneum, chest, left supraclavicular fossa commonest) → resect all measurable disease;** the authors stage infra- and supradiaphragmatic resections separately. **Histology is discordant between sites in 22–46%;** non-retroperitoneal sites show more necrosis.
- **Retroperitoneal necrosis predicts necrosis elsewhere** (only about 13% discordant) → **perform RPLND first;** with necrosis in the RPLND specimen, observing small residual masses at other sites is reasonable.
- **Viable malignancy — post-operative chemotherapy is controversial.** Pooled analysis (Fizazi): adverse factors are **IGCCCG intermediate/poor risk, incomplete resection and >10% viable cells** (5-year survival 100%, 83%, 51% for 0, 1, 2–3 factors). Post-operative chemotherapy improved relapse-free but not overall survival (apparent benefit only with one factor, not confirmed on validation). **Completeness of resection is the critical determinant;** immediate chemotherapy or surveillance depending on resection, risk group and viable fraction; no consensus regimen.
- **French BEP×3 versus EP×4 trial:** PCS was not protocol-mandated (only half had it, often mass-only); **70% of relapses and half of GCT deaths occurred in men without PCS or after inadequate RPLND** → proper integration of surgery prevents deaths.

HISTOLOGY OF RESIDUAL MASSES BY SETTING

Setting	Necrosis	Teratoma	Viable malignancy
NSGCT, retroperitoneum after first line	≈40%	≈45%	≈15%
NSGCT, non-retroperitoneal sites	≈60%	≈30%	≈10%

Setting	Necrosis	Teratoma	Viable malignancy
NSGCT, residual ≤ 20 mm (complete responders)	—	$\approx 30\%$	$\approx 6\%$
NSGCT, mediastinal primary	—	—	Up to 53%
NSGCT, after salvage chemotherapy	$\approx 26\%$	21–34%	38–53%
Seminoma, after first line	$\approx 90\%$	Rare	$\approx 10\%$

Concept — Neither chemotherapy response, imaging nor PET can distinguish necrosis from teratoma in NSGCT, so resection of every residual mass over 1 cm, retroperitoneum first, is the price of cure — and completeness of resection is the dominant prognostic factor.

Relapsed Nonseminoma Germ Cell Tumour

- Treatment depends on previous therapy and, sometimes, relapse site; **chemotherapy-naïve relapse has a far better prognosis than relapse after chemotherapy.**

Chemotherapy-naïve NSGCT relapse

- Follows surveillance or RPLND for CS I, or RPLND alone for CS IIA–B. **Markers are raised in 60–75% of surveillance relapses.**
- **Induction chemotherapy by IGCCCG group → cure >95%.** Surveillance relapse confined to the retroperitoneum, nonbulky (<3 cm) with normal markers → **induction chemotherapy or RPLND** (especially if the primary contained teratoma); RPLND $\pm$ adjuvant chemotherapy cures nearly 100% and spares chemotherapy toxicity.
- **Relapse after RPLND is usually in the lungs or mediastinum**, virtually always cured by first-line chemotherapy.
- Most relapses occur within 2 years; **relapse beyond 2 years with normal markers → biopsy or resection (teratoma likely).** Unlike post-chemotherapy relapse, **chemo-naïve relapse after 2 years carries the same prognosis as earlier relapse.**

Post-chemotherapy NSGCT relapse: early

- **Relapse within 2 years of chemotherapy = early; treated with second-line (salvage) chemotherapy.** Beyond 2 years = late relapse, which differs in prognosis and treatment.
- **Incomplete responders** — no complete response to first line, or relapse within 6 months of one — fare worst.
- Pooled analysis of about 2000 patients: **median PFS 10 months, overall survival 41 months**; adverse factors: incomplete response, **primary mediastinal NSGCT (hazard ratio 3.0)**, NPVM, raised AFP and hCG.
- **Standard second-line regimens: VIP $\times$ 4, VeIP $\times$ 4 (after prior etoposide) and TIP $\times$ 4** — never compared randomly; long-term remission 23–35% with VIP/VeIP, relapse-free survival 36–47% with TIP.
- **HDCT as second or third line:** the Indiana series (two HDCT cycles) reported 63% continuously disease-free (70% second line, 45% third line); a matched-pair analysis suggested about 10% benefit; but the **randomised trial (VIP/VeIP $\times$ 3 + one high-dose cycle vs VIP/VeIP $\times$ 4) showed no difference** in response, event-free or overall survival.
- **Why the discrepancy:** selection bias in single-arm series; expertise not reproducible elsewhere; a suboptimal trial schedule (three standard cycles, one HDCT cycle; only 73% received HDCT; toxic deaths 7% vs 3%) versus the Indiana approach (early transfer to HDCT, two cycles, 94% completion, 2.7% treatment deaths).
- **Three negative randomised trials (two first line, one second line) → HDCT is not standard; offer only at specialised centres.**

- **High-risk relapse (incomplete responders):** standard second-line or HDCT at a specialised centre; **relapse more than 6 months after first line → standard-dose second line preferred.**
- **Growing teratoma syndrome:** falling or normal markers with enlarging, usually cystic masses during chemotherapy → **interrupt chemotherapy and resect;** favourable after complete resection.
- **After second-line failure:** HDCT (if not yet given) or combinations of gemcitabine, paclitaxel, oxaliplatin and irinotecan.

Management of postsalvage chemotherapy residual masses

- **Serological complete response with residual masses → postsalvage chemotherapy surgery (PSCS).**
- Versus PCS after first line: **complete resection only 56–72% (vs ≥85%);** more viable malignancy and less necrosis and teratoma (table above); **5-year survival 44–61%; viable malignancy in PSCS specimens carries a poor prognosis not improved by post-operative chemotherapy.**

Desperation surgery

- **Rising markers despite first- and second-line chemotherapy but resectable disease at a single site (usually retroperitoneum) → salvage ("desperation") surgery.**
- Markers normalise in 47–60%; **long-term survival 33–57% (± post-operative chemotherapy);** up to half have only necrosis or teratoma.

Post-chemotherapy NSGCT relapse: late

- **Late relapse = more than 2 years after chemotherapy; about 3% of NSGCT. Biopsy to confirm,** especially with normal AFP and hCG.
- **Histology:** viable malignancy 54–88% (**yolk sac tumour commonest**), teratoma 12–28%, malignant transformation 10–20% (**adenocarcinoma commonest**).
- **Risk factors:** previous relapse; teratoma in PCS specimens (possibly incomplete resection).
- Usually a single site: **retroperitoneum 50–72%,** lungs 17%, mediastinum 9%, neck 7%, pelvis 4% → **failure to control the retroperitoneum initially is a major risk factor.**
- **AFP raised in about half, hCG in about a quarter; marker-only relapse → monitor until measurable disease appears.**
- **Chemoresistant; outcome depends on complete surgical resection** (teratoma and transformation are chemoinensitive, viable disease is platinum-resistant). Contemporary data suggest cure rates may approach those of early relapse. Chemotherapy rarely achieves complete response, whereas initial surgery renders most men disease-free; complete resection roughly doubles cancer-specific survival; symptoms and multifocal disease predict worse survival.
- **TIP gives the best chemotherapy results; aggressive surgery to resect all disease — primarily, or after chemotherapy if initially unresectable.**

NSGCT RELAPSE CATEGORIES COMPARED

	Chemo-naive	Early post-chemotherapy (≤2 years)	Late post-chemotherapy (>2 years)
Arises after	Surveillance or RPLND	First-line chemotherapy	Chemotherapy (≈3%)
Typical site	Retroperitoneum (surveillance); lung/mediastinum (after RPLND)	Variable	Retroperitoneum 50–72%, single site
Treatment	Induction chemotherapy by IGCCCG; RPLND if nonbulky, marker-negative	Second-line: VIP, VeIP, TIP; HDCT at specialised centres	Complete surgical resection; TIP if chemotherapy needed
Prognosis	Cure >95%; timing irrelevant	Median survival 41 months; incomplete responders worst	Depends on resection; may approach early relapse

KEY POINTS — NSGCT

- CS I: surveillance, primary RPLND and primary BEP chemotherapy are all accepted, with survival approaching 100%.
- Full bilateral nerve-sparing template for suspicious nodes; modified template when nodes are clinically negative. Preserving ejaculation must not compromise oncological efficacy; experienced surgeons only.
- CS IIA–B: induction chemotherapy or primary RPLND; long-term cure $\geq 95\%$.
- CS IS, IIC and III: cisplatin-based induction, regimen and cycles by IGCCCG.
- Resect all post-chemotherapy residual masses: residual cancer (viable malignancy or teratoma) in 50% or more.

Concept — Relapse prognosis tracks previous chemotherapy exposure: chemo-naïve relapse is simply first-line treatment delayed, early post-chemotherapy relapse tests platinum resistance, and late relapse is a surgical disease of teratoma, transformation and platinum-resistant yolk sac tumour.

Seminoma**Clinical Stage I Seminoma**

- **About 80% of seminomas are CS I — the commonest presentation of testis cancer.**
- **Accepted options: surveillance, primary radiotherapy and single-agent carboplatin**, each with long-term cancer control approaching 100%, largely validated by randomised trials.
- **Current aim is reducing therapeutic burden:** platinum and infradiaphragmatic radiotherapy raise late cardiovascular and SMN risk; radiotherapy fields and doses have been cut; carboplatin is less neuro-, oto- and nephrotoxic than cisplatin, but its late cardiovascular and SMN risks are largely unknown.

Primary radiotherapy

- **Historical mainstay: retroperitoneum plus ipsilateral pelvis ("dog-leg", DL, field);** optimal dose undefined — **most centres used 25–35 Gy in 15–20 fractions.** Cancer-specific survival $\approx 100\%$; progression-free 95–97%.
- **In-field recurrence $<1\%$ → no routine abdominal–pelvic CT.** Inguinal metastasis uncommon without previous inguinal or scrotal surgery. **Relapse typically in the thorax and left supraclavicular fossa**, virtually always cured by first-line chemotherapy; isolated inguinal relapse → radiotherapy or excision.
- **Follow-up after DL radiotherapy:** clinical assessment, chest radiograph, markers.
- **Acute toxicity:** mild self-limiting nausea, vomiting and diarrhoea; grade II–IV haematological toxicity 5–15%.
- **Late toxicity:** moderate and severe late gastrointestinal toxicity (dyspepsia, peptic ulcer) in 5% and $<2\%$; **scatter to the contralateral testis despite shielding → persistent oligospermia in 8%; SMN risk about 18% at 25 years** — leukaemia with radiotherapy and chemotherapy; upper gastrointestinal, bladder and possibly pancreatic cancer with radiotherapy.
- **Shrinking field: DL → para-aortic (PA) strip**, because pelvic nodes are involved in only 1–3% without previous inguinal or scrotal surgery; PA may reduce SMN and speeds early recovery of spermatogenesis, but its small **pelvic recurrence risk mandates routine pelvic CT**, with its cost and radiation.
- **Shrinking dose: 20 Gy matches 30 Gy** with less acute gastrointestinal toxicity, leukopenia and lethargy (equal by 12 weeks); durability still under follow-up. Trial results in the table below (EORTC, European Organisation for Research and Treatment of Cancer).

Surveillance

- **Rationale:** late toxicity of DL radiotherapy, **80–85% cure by orchidectomy**, and >90% chemotherapy cure of advanced seminoma.
- **Harder than NSGCT surveillance: markers rarely detect relapse**, and **10–20% of relapses occur 4 years or more after diagnosis** → long-term CT. About 60% of contemporary US CS I seminoma patients are under surveillance.
- **5-year relapse-free survival 80–86%; cancer-specific survival ≈100%. 84–100% of relapses are retroperitoneal**; 18–24% have bulky or distant disease at relapse. **DL radiotherapy treats 73–88% of relapses, curing 70–90%**; relapses outside the retroperitoneum are cured by first-line chemotherapy.
- **Median time to relapse: seminoma 14 months versus NSGCT 4–8 months**; more seminoma relapses occur after 5 years.
- **Schedule:** assessment (examination, chest radiograph, markers, CT) every 2–4 months in years 1–3, 6-monthly in years 4–7, then annually; CT every 4–6 months in years 1–3, 6-monthly in years 4–7, then annually. The MRC evidence for fewer scans comes from low-risk NSGCT; **its applicability to seminoma is unclear. Long-term follow-up is mandatory.**
- **Detection: CT finds >85% of relapses, hCG <5%, chest radiograph and examination virtually none.**
- **Risk factors (pooled 1980s surveillance series): tumour >4 cm and rete testis invasion; LVI is not predictive in seminoma (unlike NSGCT).** 5-year relapse 12%, 16%, 32% with 0, 1, 2 factors; about one-fifth had both. Treating all "high-risk" patients would still overtreat two-thirds; **not prospectively validated.**

Primary chemotherapy with single-agent carboplatin

- **One or two cycles;** rationale: complete response in 65–90% of advanced seminoma and less toxicity than cisplatin. Published series: no seminoma deaths; 3–5-year relapse-free rates 91–100%.
- **MRC TE19/EORTC 30982 (1477 men):** one cycle of carboplatin equivalent to PA radiotherapy (table below), with less lethargy and time off work; grade III–IV haematological toxicity 4%.
- **Relapse after carboplatin is salvageable** (pooled analysis: 5-year overall survival 98%), not unduly chemoresistant; **15% of relapses occur after 3 years → at least 5 years of surveillance.**
- **Underdosing concern:** relapse is higher with one than two cycles and with inadequate dosing. A Swedish–Norwegian Testicular Cancer Group (SWENOTECA) study found one dose only about halved relapse in men with risk factors (tumour >4 cm: roughly 19% → 10%; rete testis invasion: roughly 14% → 9%); a German study likewise found relapse 5% after one cycle versus 1.5% after two. **Guidelines are split on one versus two cycles**, and some question carboplatin for stage I altogether.
- **Calvert formula: carboplatin dose (mg) = 7 × (glomerular filtration rate [GFR, mL/min] + 25); never use estimated GFR** → base a single cycle on isotope GFR, or give two cycles.
- **Guidelines prefer surveillance** (low overall relapse risk, no validated high-risk markers, late toxicity of alternatives): 80–85% avoid treatment and relapses are mostly salvaged with DL radiotherapy; surveillance must run beyond 5 years with frequent CT. **Non-compliant or unwilling patients → radiotherapy or carboplatin ×2.**

CLINICAL STAGE I SEMINOMA — SURVEILLANCE VERSUS RADIOTHERAPY VERSUS CARBOPLATIN

	Surveillance	Radiotherapy (PA 20 Gy or DL)	Carboplatin
Relapse	14–20% (5-year relapse-free 80–86%)	3–5% (progression-free 95–97%)	≈5% (one cycle), ≈1.5% (two cycles)
Relapse site	Retroperitoneum (84–100%)	Thorax, left supraclavicular fossa; pelvis after PA field	Retroperitoneum
Imaging follow-up	Long-term abdominal CT	Chest radiograph (DL); pelvic CT (PA)	≥5 years

	Surveillance	Radiotherapy (PA 20 Gy or DL)	Carboplatin
Main drawback	Late relapses; CT burden; compliance	SMN, gastrointestinal toxicity, oligospermia	Underdosing; unknown late effects
Status	Preferred	For non-compliant/unwilling	×2 for non-compliant/unwilling

CLINICAL STAGE I SEMINOMA — RANDOMISED TRIALS

Trial	Comparison	Endpoint result	Verdict
MRC	DL vs PA radiotherapy	Equal relapse-free and overall survival; pelvic relapse 2% vs 0% with PA	PA acceptable, needs pelvic CT
MRC/EORTC	20 vs 30 Gy PA	Equal 5-year relapse-free survival; less acute toxicity at 20 Gy	20 Gy
MRC TE19/ EORTC 30982	Carboplatin ×1 vs PA 20–30 Gy	5-year relapse-free 94.7% vs 96%	Carboplatin equivalent

- **Trap:** in seminoma, **rete testis invasion and size >4 cm** are the risk factors — not LVI, which is the NSGCT predictor.

Concept — CS I seminoma is the purest case for de-escalation: occult disease is only 15–20%, relapses are late but almost always retroperitoneal and salvageable, so surveillance wins — and the residual debate concerns late toxicity (radiotherapy) and adequate dosing (carboplatin).

Clinical Stage IIA and IIB Seminoma

- **15–20% of seminomas are CS II, 70% of them IIA–B.**
- **DL radiotherapy 25–30 Gy, including a 5–10 Gy boost to involved nodes:** well tolerated (acute grade III–IV gastrointestinal toxicity 8–10%); **prophylactic left supraclavicular radiotherapy abandoned** (<3% would benefit). In-field recurrence 0–2% (IIA) and 0–7% (IIB); disease-free survival in the table below. Adding carboplatin cut relapse in one series (needs confirmation). Relapses virtually all cured by chemotherapy; **no routine CT after complete resolution.**
- **Induction chemotherapy (BEP×3 or EP×4) is an accepted alternative:** Spanish series — complete response 83%, residual masses rare and necrotic; SWENOTECA — no relapses after chemotherapy versus 10% after radiotherapy. **Preferred for bulky (>3 cm) and/or multiple retroperitoneal masses**, where relapse after radiotherapy is higher.
- **Primary RPLND:** long standard for marker-negative CS IIA–B NSGCT but not for seminoma, for historical reasons; interest has revived given chemotherapy and radiotherapy late toxicity and because **disease is retroperitoneum-confined in >90% of IIA and >85% of IIB.**
- **Surgery in Early Metastatic Seminoma (SEMS) trial:** long-term complications 7%; recurrences salvaged by chemotherapy or further surgery. **COTRIMS trial:** relapse about 10%. → **RPLND is an appropriate option for CS II seminoma with nodes ≤3 cm**, sparing most men chemotherapy.

CLINICAL STAGE IIA–B SEMINOMA — TREATMENT OPTIONS

	DL radiotherapy	Induction chemotherapy	Primary RPLND
Regimen	25–30 Gy with 5–10 Gy boost	BEP×3 or EP×4	Template dissection
Best suited	Nonbulky (≤3 cm)	Bulky >3 cm or multiple masses	Nodes ≤3 cm
Disease control	DFS 92–100% IIA; 87–90% IIB	Relapse-free ≈90%; overall ≈95%	2-year recurrence-free ≈81% (SEMS); overall 100%

	DL radiotherapy	Induction chemotherapy	Primary RPLND
Drawback	Late radiation toxicity; more relapse with bulk	Late chemotherapy toxicity	Evidence still maturing

Concept — Bulk decides modality: nonbulky retroperitoneal seminoma can be treated locally (radiotherapy or, increasingly, RPLND), whereas bulky or multifocal nodes need systemic chemotherapy.

Clinical Stage IIC and III Seminoma

- **Induction chemotherapy by IGCCCG group**, as for NSGCT.
- **90% are good risk** → BEP×3 or EP×4: complete radiographic response 70–90%; 5-year overall survival 91% in the series quoted.
- **10% are intermediate risk (NPVM)** → BEP×4.
- **Single-agent carboplatin is inferior to cisplatin regimens** in advanced seminoma.

Management of post-chemotherapy residual masses

- **Residual masses in 58–80%; spontaneous resolution in 50–66% (30–50% if >3 cm), median 12–18 months.**
- **Histology: necrosis about 90%, viable malignancy about 10%.**
- **PCS is technically difficult and often unfeasible — post-chemotherapy desmoplastic reaction:** complete resection only 58–74% (vs ≥85% in NSGCT); adjunctive procedures such as nephrectomy in 38–51%.
- **Teratoma and transformation are much less of a concern; post-chemotherapy radiotherapy has no role.**
- **Size predicts viable disease: discrete masses >3 cm contain viable malignancy in 13–55%, <3 cm in 0–4%.**
- **FDG-PET is useful in seminoma:** prospectively, for masses >3 cm, **specificity 100%, sensitivity 80%**; later series report false positives in 22–78%, but **a negative PET remains highly accurate**. False positives are commoner within 4–6 weeks (inflammation, non-viable tumour) → **PET no earlier than 6 weeks after chemotherapy.**
- **Consensus: discrete residual mass >3 cm → PET at ≥6 weeks; PET-negative masses and masses <3 cm → observe.**
- **PET-positive mass >3 cm is controversial:** repeat PET at 6–8 weeks (unproven); or observation with delayed intervention for growth (only 14–20% relapse, mostly within 4–6 months) — but in the Indiana experience delayed intervention achieved durable control in few, with 72% relapse after post-chemotherapy RPLND (PC-RPLND). Up-front PC-RPLND assesses response immediately and allows timely further chemotherapy; teratoma is present in up to one-third of residual masses >3 cm; completely resected viable seminoma stayed disease-free whereas unresected viable disease was often fatal. → **Strongly consider PC-RPLND for PET-positive masses >3 cm, especially after second-line chemotherapy.**
- **Mass-only resection → 33% local relapse** → perform a full bilateral or accepted modified template where technically feasible.

POST-CHEMOTHERAPY RESIDUAL MASSES — NSGCT VERSUS SEMINOMA

	NSGCT	Seminoma
Commonest histology	Teratoma ≈45%, necrosis ≈40%, viable ≈15%	Necrosis ≈90%, viable ≈10%
Threshold for action	Any mass >1 cm → PCS	>3 cm → PET at ≥6 weeks; <3 cm observe
FDG-PET	No role (teratoma not FDG-avid)	Useful; negative PET reliable
Resectability	Complete resection ≥85%	58–74% (desmoplasia); nephrectomy often needed

	NSGCT	Seminoma
Spontaneous resolution	Not relied on	50–66% over 12–18 months
Extent of surgery	Full bilateral template	Template preferred (mass-only → 33% local relapse)

Concept — In seminoma the residual mass is usually necrosis and rarely teratoma, so — unlike NSGCT — size and PET, not routine resection, select the few patients who need difficult post-chemotherapy surgery.

Relapsed Seminoma

Chemotherapy-naïve seminoma relapse

- Follows surveillance (CS I) or primary radiotherapy (CS I–IIB).
- **After surveillance: DL radiotherapy for 73–88%, cure 70–90%; bulky (>3 cm) retroperitoneal or systemic relapse → first-line chemotherapy, salvage ≈100%.**
- **Relapse outside the retroperitoneum after radiotherapy → first-line chemotherapy, virtually always curative.**
- **Nonbulky (≤3 cm) retroperitoneal relapse → RPLND as an alternative to radiotherapy or chemotherapy.**
- **Relapse after single-agent carboplatin counts as chemo-naïve → first-line cisplatin-based chemotherapy.**

Post-chemotherapy seminoma relapse: early

- **15–20% of advanced seminoma relapse after induction** (10% after an initial complete response); incomplete responders and relapse after a major response → long-term survival 20–50%.
- Regimens are borrowed from NSGCT: **VeIP×4** — complete response in most, with about half continuously free of recurrence long term; **HDCT for incomplete responders** — about half remained disease-free.
- **Normal markers at relapse → biopsy before second-line chemotherapy**, because teratoma (transformed NSGCT) may be present.

Post-chemotherapy seminoma relapse: late

- **Pure seminoma accounts for under 8% of late relapses** in most series; one large series found 41%, mostly after carboplatin or radiotherapy rather than cisplatin, with long-term control in 88% → **late seminoma relapse may carry a favourable prognosis, especially without previous cisplatin.**

KEY POINTS — SEMINOMA

- CS I: surveillance, primary radiotherapy (20–30 Gy to the PA region ± ipsilateral pelvis) or carboplatin, each with survival approaching 100%.
- Given the low risk of occult metastasis, surveillance is the recommended approach for CS I.
- CS IIA–B nonbulky (<3 cm): DL radiotherapy, first-line chemotherapy (BEP×3 or EP×4) or RPLND; bulky (>3 cm) and/or multifocal nodes → first-line chemotherapy.
- CS IIC and III: cisplatin-based chemotherapy, regimen and cycles by IGCCCG.
- Discrete residual masses >3 cm → FDG-PET; PET-positive → resection; PET-negative or <3 cm → safe observation.

Brain Metastases

- **About 1% of disseminated GCT have brain metastases before chemotherapy; 0.4–3% develop them after first-line chemotherapy.**
- **Associated with choriocarcinoma — suspect whenever hCG is very high.**

- **Choriocarcinoma is highly vascular and bleeds during chemotherapy: intracranial haemorrhage deaths 4–10% → evaluate neurological change urgently.**
- **5-year survival with brain metastases: NSGCT 33%, seminoma 57%.**
- **Brain relapse after complete response is worse than brain metastasis at diagnosis:** survival 39–44% if isolated, 2–26% with other sites.
- No randomised trials; most receive multimodal therapy (chemotherapy, surgery, whole-brain radiotherapy, stereotactic radiosurgery).
- **At diagnosis: BEP×4, then resection of residual lesions;** benefit of radiotherapy unclear — the authors reserve it for unresectable residual lesions unsuitable for radiosurgery, because of radiation neurotoxicity.
- **Brain relapse after first-line chemotherapy: second-line chemotherapy, then resection and/or radiotherapy.**

KEY POINTS — BRAIN METASTASES

- Brain metastases are associated with choriocarcinoma; suspect them with a very high hCG.
- Choriocarcinomas are highly vascular and haemorrhage during chemotherapy, causing intracranial haemorrhage.

Concept — Brain metastasis is a choriocarcinoma problem: hCG identifies the risk, haemorrhage is the immediate danger, and relapse in the brain after complete response signals resistant disease with a worse outlook than brain involvement at diagnosis.

Treatment Sequelae

- Divided into early and late. **Main problems after RPLND: midline scar, ejaculatory dysfunction, small-bowel obstruction and perioperative complications; hypogonadism is increased after orchidectomy.**

Early Toxicity

- **Cisplatin-based chemotherapy:** fatigue, myelosuppression, infection, peripheral neuropathy, hearing loss, reduced renal function, death.
- **Toxic death rate: 0–2.4% in good-risk and 3–4.4% in intermediate/poor-risk first-line chemotherapy.**
- **Most men father children after treatment, but paternity is lower after radiotherapy and/or chemotherapy.**
- **Early radiotherapy effects:** fatigue, nausea and vomiting, leukopenia, dyspepsia.

Late Toxicity

LATE EFFECTS IN GCT SURVIVORS

Effect	Frequency
Raynaud phenomenon	20–45%
Peripheral neuropathy	14–43%
Hearing loss / tinnitus after cisplatin	20–40% symptomatic; 30–75% on audiometry
Hypogonadism	10–20% orchidectomy alone; 15–40% radiotherapy; 20–25% first-line chemotherapy
Non-GCT second cancers	60–100% excess after cisplatin or radiotherapy; 200% excess after both; SMN mortality roughly doubled

- Other late sequelae: infertility, SMN, **cardiovascular disease.**

- **Non-cancer mortality:** radiotherapy → gastrointestinal and cardiovascular deaths; chemotherapy → infection, cardiovascular and pulmonary deaths; **combined radiotherapy and chemotherapy → highest risk.**
- **Cardiovascular disease is the best-documented late effect;** putative causes are radiation- or chemotherapy-induced vascular injury, chemotherapy-induced cardiac injury and metabolic syndrome.
- **Surveillance CT adds radiation** that may itself raise SMN risk.

KEY POINTS — TREATMENT SEQUELAE

- All GCT treatments (surgery, radiotherapy, chemotherapy) carry early and late toxicity.
- The most concerning late complications are cardiovascular disease and secondary malignancy.
- As most patients are cured, even with advanced disease, minimising toxicity without compromising cure is a key objective.

Concept — Survivorship is now the deciding endpoint: because cure is expected, late cardiovascular disease and second cancers — rising with the number of modalities used and with cumulative radiation, including surveillance CT — decide between otherwise equally curative options.